



MT1003 – Calculus and Analytical Geometry

Assignment No: 01

Individual Assignment

Section: All BSCS Sections

Semester: Fall 2023

Due date: Thursday, 14 September 2023, 01:00pm

Marks: 16*10=160

Instructions:

1. Plagiarized work will result in zero marks.
 2. No retake or late submission will be accepted.
 3. Attach complete code, results, and screenshot for questions that require programming solution. Programs/codes should be typed.
 4. Questions with  icon require partial or complete solution using MATLAB.
 5. **The assignment is to be submitted in softcopy as well as in hardcopy. Submit the hardcopy before the deadline through CR, and softcopy on GCR.**
 6. The softcopy should be a single PDF file of your complete assignment including programming and non-programming questions.
 7. The PDF file should be according to the following format: id_section_A1 e.g., i23-123456_A_A1. A1 in the end denotes Assignment 1. **The first page must include complete student information, including name, section, id, course name, and assignment number.**
 8. The images of by-hand solution should be properly scanned. You can use any mobile application such as CamScanner or Adobe Scan for scanning. Each of these applications allow you to export pdf or image files which you can use to combine with your programming solutions. Do not attach direct images from the camera application of your mobile phone, or screenshots.
 9. MATLAB is the only approved programming tool. Octave-Online is also allowed.
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Question 1

You are designing a rectangular garden next to a wall. The garden has two vertical sides, one of which is the wall, and a flat top. The flat top is made of concrete that costs \$8 per square foot, and the other two sides are made of wooden fencing, which costs \$4 per linear foot. You have a budget of \$800 for the garden's construction. Express the width 'w' as a function of the length 'l' of the garden. Also find the domain of the function.

Question 2



Taxicabs such as Uber and Lyft have a unique fare structure. Passengers are charged a flat fee of \$3.00 for entering a cab and then an additional \$0.75 for each quarter of a mile (or fraction thereof) traveled. Ignoring any additional charges, if 'x' represents the distance traveled in miles, write a function $C(x)$ that represents the cost of the taxi fare.

Using MATLAB, sketch the graph of $C(x)$. Then, based on the graph of the fare function, find each of the following limits, if they exist:

- | | | | |
|------|---------------------------------------|---------------------------------------|-------------------------------------|
| i) | a) $\lim_{x \rightarrow 0.30^-} C(x)$ | b) $\lim_{x \rightarrow 0.30^+} C(x)$ | c) $\lim_{x \rightarrow 0.30} C(x)$ |
| ii) | a) $\lim_{x \rightarrow 0.45^-} C(x)$ | b) $\lim_{x \rightarrow 0.45^+} C(x)$ | c) $\lim_{x \rightarrow 0.45} C(x)$ |
| iii) | a) $\lim_{x \rightarrow 0.75^-} C(x)$ | b) $\lim_{x \rightarrow 0.75^+} C(x)$ | c) $\lim_{x \rightarrow 0.75} C(x)$ |

Question 3

You are an automotive engineer developing a fuel-efficient hybrid car, the "EcoDrive 2024." The car's MPG (miles per gallon) varies with speed according to the following piecewise function:

$$\begin{cases} 100 - s & \text{if } s \leq 55 \text{ mph} \\ \frac{2200}{s} & \text{if } s > 55 \text{ mph} \end{cases}$$

- Find the speed for maximum MPG in electric mode and with the gasoline engine.
- Calculate the MPG at specific speeds like 45 mph, 60 mph, and 70 mph.
- Determine the speed at which the car transitions from electric to gasoline mode.
- Discuss whether the MPG function is continuous at the transition speed (55 mph) and when approaching it from different directions. Provide reasons for your conclusions.

Question 4

Use the graph of the function f to decide whether the value of the given quantity exists. If it does, find it. If not, explain why.

- $\lim_{x \rightarrow 10} f(x)$
- $\lim_{x \rightarrow -2^+} f(x)$



- c. $\lim_{x \rightarrow -8} f(x)$
- d. $\lim_{x \rightarrow 6} f(x)$
- e. $f(-8)$
- f. the vertical asymptotes of the graph of f

Question 5

Classify the locations and types of discontinuities of the following functions.

- a. $f(x) = \frac{x^2}{x^2-1}$
- b. $f(x) = \text{greatest integer function}$
- c. $f(x) = \frac{\sin(\pi x)}{x}$
- d. $f(x) = \begin{cases} x^2 - 2 & \text{if } x \neq -3 \\ 5 & \text{if } x = -3 \end{cases}$
- e. $f(x) = \begin{cases} e^x & x < 0 \\ x^2 & x \geq 0 \end{cases}$

Question 6

Evaluate $\lim_{x \rightarrow 1} \arccos\left(\frac{1-\sqrt{x}}{1-x}\right)$.

Question 7

Use the Squeeze Theorem to find the limit of $\lim_{x \rightarrow -\infty} \frac{5x^2 - \sin(3x)}{x^2 + 10}$.

Question 8

A new pizzeria has opened near your home, the owner has decided to offer a 20% discount on the opening day on the pizza whose original price is x dollars. The owner is offering additional 15% discount coupon to those who agree to fill a feedback form afterwards. You and your friends have agreed to provide feedback and thus have become entitled to the additional 15% discount coupon. Use the concept of composite functions to identify the amount that you will pay after applying the coupon on the sale price.

Question 9

Scientists have developed a radioisotope-based power supply whose power output in watts is given by the following function:

$$P = 75e^{-\frac{t}{125}}$$

where t is the number of days the power supply is used. The experimental power supply has been installed in an unmanned undersea vehicle on an exploration mission. The vehicle requires 7watts to operate at full capacity. Use the given information to find the number of days for which the vehicle can operate at full capacity.



Question 10



The function $H(t) = 8\sin\left(\frac{\pi}{6}t\right)$ models the height H (in feet) of the tide t hours after midnight. Assume that $t = 0$ is midnight.

- Find the amplitude and period.
- Graph the function over one period by hand.
- Graph the function using MATLAB.
- What is the height of the tide at 4:30 a.m.?

Question 11



The local mass transit authority has given you the following passenger data to consider:

Year	1999	2000	2001	2002	2003	2004	2005	2006
Passengers (in millions)	76.4	78.6	81.5	87.8	88.4	92.4	94.0	95.0

Use MATLAB to plot the data and observe its trend. Find a linear function for the data and use it to predict the number of passengers for 2014.

Question 12



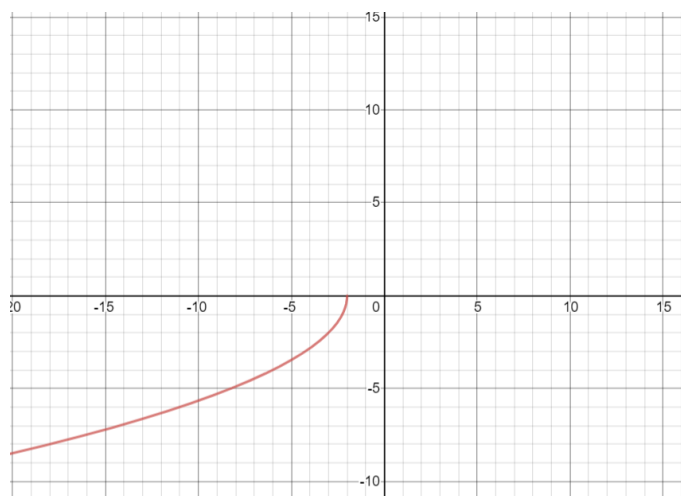
Sketch the graphs of the functions, not by plotting points but by starting from basic function and applying transformations. Also plot the graphs using MATLAB and verify your answers.

- $y = \frac{1}{2}e^{-x} - 1$
- $y = \ln(x - 2) - 1$
- $y = 2 + \sqrt{1 - (x - 1)^2}$

Question 13



Identify the base graph and all the transformations required to obtain the given graph from base graph. Verify the identified transformations using MATLAB. In the end you should obtain the function that will result in the given graph.



Question 14

Identify the domain of the following functions.

a) $f(x) = \sin^{-1}(3x - 1)$

b) $j(x) = \frac{1}{x - \sqrt{x+2}}$

c) $f(x) = \ln|x + 3| - 5$

Question 15



Graph $y = \tan x$ and $y = \cot x$ together for $-7 \leq x \leq 7$. Comment on the behavior of $\cot x$ in relation to the signs and values of $\tan x$.

Question 16



Graph $y = \ln |\sin x|$ in the window $0 \leq x \leq 22$, $-2 \leq y \leq 0$. Explain what you see. How could you change the formula to turn the arches upside down?